

Types of LLMs

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My folder: C:\Users\ashok\OneDrive\Documents\llm

One reference is [here](#) and another one 'A comprehensive review of LLMs' is here.

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Summary of LLM models

TOP 7 TYPES OF LLMs

Understanding Large Language Model Categories in 2025

1 🧠 **Base Models**

What it is:
Trained on massive unlabeled datasets, they're brilliant at text prediction but not inherently great at instruction following.

Key example:
Llama, Mistral, and other foundation models that serve as building blocks for more specialized systems.

2 🎯 **Instruction-Tuned Models**

What it is:
Base models that are fine-tuned with human feedback to align with user intent.

Key example:
ChatGPT and Claude - designed to follow instructions and be helpful, harmless, and honest.

3 🧠 **Reasoning Models**

What it is:
Trained to write their thought process before giving the final answer. It significantly improves model's accuracy.

Key example:
Claude 3.7 Sonnet with reasoning mode enabled shows step-by-step problem-solving.

4 🏢 **Mixture of Experts (MoE)**

What it is:
A clever architectural twist that allows models to scale to trillions of parameters without breaking compute budgets, by activating only the relevant "experts" per task.

Key example:
Qwen3-235B-A22B — third-gen model, 235B total parameters, only 22B active per token via MoE (8 of 128 experts active at a time).

5 🖼️ **Multimodal Models (MLLMs)**

What it is:
These models process images, audio, and text together — enabling AI to reason over, extract information and answer questions about visual and audio inputs.

Key example:
GPT-4o, Claude 3 Opus, and Gemini can analyze images and respond to visual prompts.

6 ⚡ **Hybrid Models**

What it is:
Dynamically decide whether a prompt needs fast execution or deeper reasoning.

Tip to use:
Remember to include "no_think" in your system prompt if you don't want model to spend time "thinking".

7 🔍 **Deep Research Agents**

What it is:
Agents that simulate the work of a human researcher: planning, browsing the web, synthesizing info, and generating structured, detailed reports.

Key example:
Claude with web search and research mode can investigate topics across multiple sources.

Get more AI insights at MADAILab.com

Exercise for students: Get list of example models from ollama library and HuggingFace

1. Foundation or Base models

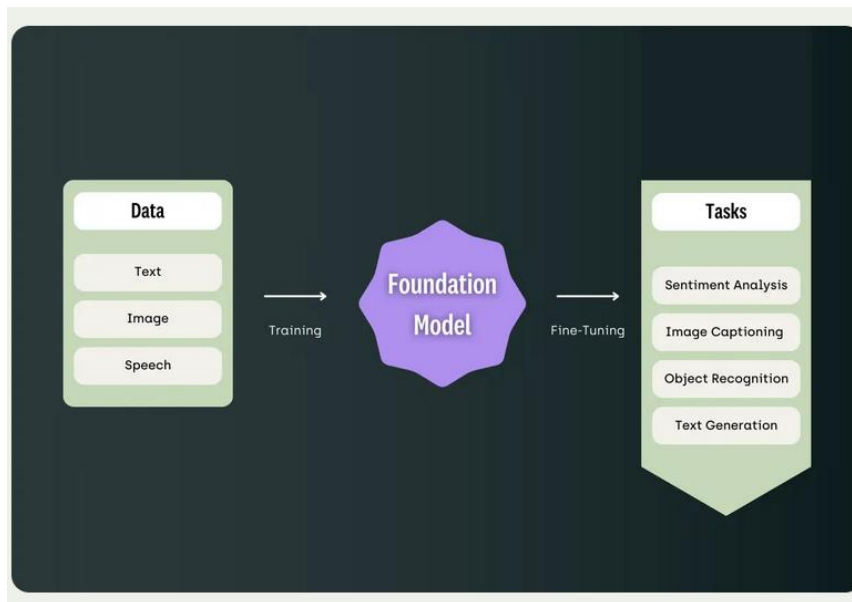
All generative LLMs are designed to take in some text and then predict what text is most likely to come after it.

"Base" models are trained on a wide variety of different texts, so they make minimal assumptions about the structure of the text they're completing.

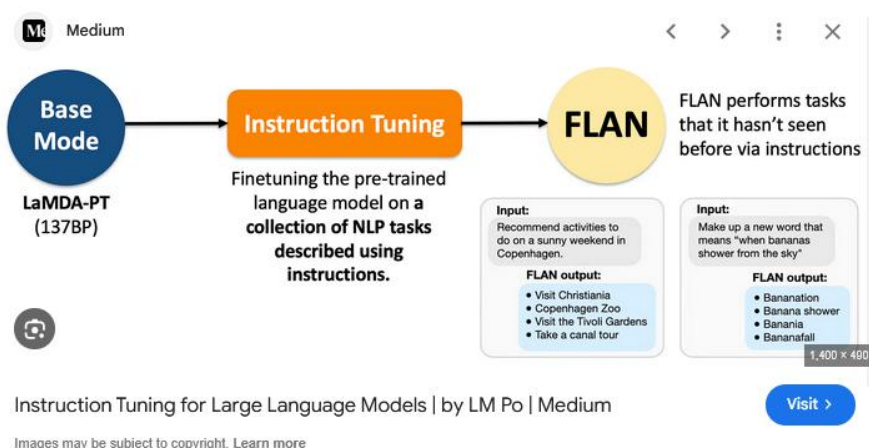
Chat models are created from base models by training them on transcripts of dialogue, so they assume that whatever text they are given is a fragment of dialogue. You can chat with them by filling in one side of a conversation and letting the model fill in the other.

Instruct models are trained on instruction–response pairs, so if you give it an instruction, the model assumes that it should continue with a response that obeys the instruction.

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2. Instruction Tuned models



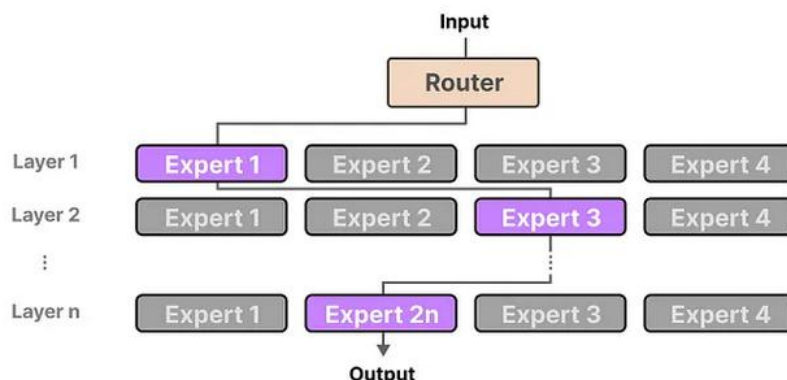
A typical dataset for instruction fine-tuning consists a) Instruction, b) Input Query and c) Output Response.

Instruction: "Translate the following English sentence to French."

Input: "The weather is nice today."

Output: "Le temps est agréable aujourd'hui."

3. Mixture of experts model



What is MOE model:

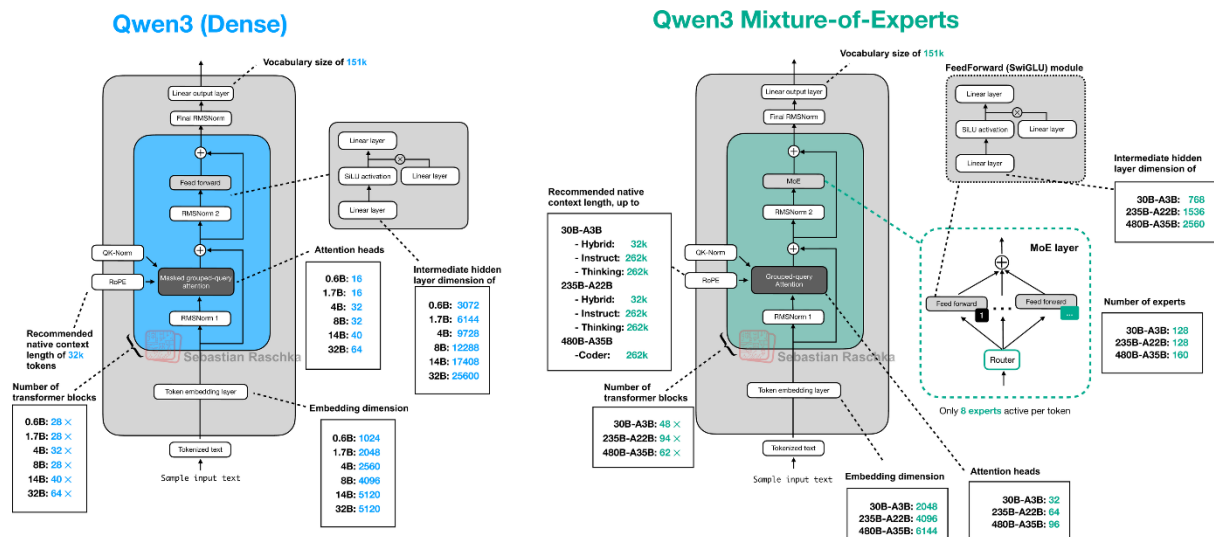
An LLM is made of a pile of “layers”, each layer having both “attention heads” (which are responsible for understanding the relationship between words in the “context window”) and a “feed forward” block (a fully connected “multi-layer perceptron”, the most basic neural network), the later part is responsible for the LLM ability to store “knowledge” and represents the majority of the parameters.

MoE just comes from the realization that you don't need to activate the whole Feed forward block for every layer at all time, that you can split every feed forward blocks in multiple chunks (called the “experts”) and that you can have a small “router” in front of it to select one or several experts to be activated for each tokens instead of activating all of them. Thus, very simply, each expert is in fact, each Feed Forward layer.

The massively reduces the computation and memory bandwidth required to power the network while keeping its knowledge storage big.

Oh ,also what kind of knowledge is stored by each “expert” is unknown and there's no reason to believe that they are actually specialized for one particular task in a way that a human expert is.

And here is a workflow for MOE as compared to foundation models.



4. Reasoning models

Reasoning models are trained by transitioning from predicting words to learning a step-by-step thinking methodology. While standard Large Language Models (LLMs) rely on massive data imitation, reasoning models are built using **large-scale Reinforcement Learning (RL) on verifiable problems**.

The process begins with a foundation or instruction-tuned model. AI developers use **Supervised Fine-Tuning (SFT)** by feeding the model a highly curated dataset filled with complex coding, math, and logic problems. Crucially, these examples don't just contain the correct final answer—they explicitly outline **detailed, step-by-step thinking traces**. This teaches the model the basic syntax of "thinking" before answering.

Example Dataset Instance: Math Word Problem

1. The Prompt (The Input)

text

"A farmer has chickens and rabbits. There are 10 heads and 28 legs in total. How many chickens and how many rabbits does the farmer have?"

Use code with caution.



2. The Target Response (The Gold Standard Output)

xml

<thought>

1. Identify the variables: Let C = number of chickens, R = number of rabbits.
2. Set up the first equation based on heads: $C + R = 10$.
3. Set up the second equation based on legs: Chickens have 2 legs, rabbits have 4 legs. $2C + 4R = 28$.
4. Try to solve by substitution. From equation 1, $C = 10 - R$.
5. Substitute C into equation 2: $2(10 - R) + 4R = 28$.
6. Expand the equation: $20 - 2R + 4R = 28$.
7. Simplify: $20 + 2R = 28$.
8. Subtract 20 from both sides: $2R = 8$.
9. Divide by 2: $R = 4$. So there are 4 rabbits.
10. Find chickens: $C = 10 - 4 = 6$.
11. Double-check the math:
 - Heads: $6 + 4 = 10$. (Correct)
 - Legs: $(6 * 2) + (4 * 4) = 12 + 16 = 28$. (Correct)
12. Formulate the final user-facing response.

</thought>

The farmer has 6 chickens and 4 rabbits.

5. Multimodal models

Multimodal models are artificial intelligence systems designed to process, understand, and reason across different types of data—or modalities—simultaneously. Instead of only handling text like traditional chatbots, they can natively analyze combinations of text, images, audio, video, and numerical data to produce context-rich outputs.  Google Cloud +3



How They Work

Unlike older systems that relied on separate AI tools (like one to "read" an image and another to describe it), multimodal models combine this processing into a single, unified architecture.  Micron Technology +1

- **Encoding:** Specialized encoders translate different data types (e.g., pixel grids for images, audio waveforms) into a shared "language" of numerical vectors.  IBM +1
- **Shared Vector Space:** Because different data types are converted into the same high-dimensional space, the model can inherently connect them (e.g., understanding that the spoken word "apple," a photo of an apple, and the written text "apple" all represent the same concept).  YouTube · AssemblyAI +3
- **Unified Reasoning:** The AI uses its central engine to relate the inputs, infer hidden patterns, and generate responses—which can also be multimodal.  IBM +1

Common Use Cases

The ability to combine data types unlocks dynamic real-world applications: 

- **Visual Question Answering:** You can upload a photo of a broken appliance, ask the AI to identify the problem, and get text instructions on how to fix it.  IBM +3
- **Data Interpretation:** The models can analyze charts, graphs, and financial reports simultaneously, outputting summarized insights.  IBM +3
- **Audio/Speech Conversations:** You can talk directly to an AI assistant, and it will respond in natural, expressive spoken audio by understanding your tone, pacing, and words.  IBM +1

MLLMs expand upon the foundations laid by traditional LLMs. While they often leverage a powerful pre-trained LLM as their backbone for language understanding and reasoning, they incorporate additional components to handle non-textual data. The core architectural difference lies in the need for specialized **encoders** for each modality. For instance, an MLLM might use a Vision Transformer (ViT) or a Convolutional Neural Network (CNN) to process images, an audio encoder for sound, and the standard LLM tokenizer/embedding layer for text. These encoders transform the input from each modality into vector representation. A crucial step in MLLM architecture is **embedding alignment and fusion**. The embeddings generated by the different modality encoders need to be projected into a shared space where the model can understand the relationships between them. A dedicated **fusion module** or specific training techniques (like contrastive learning) are employed to integrate these diverse representations into a unified multimodal vector space. This allows the model, for example, to connect the word "dog" in a text caption to the visual features of a dog in an accompanying image.

6. Deep Research Agent

What are Deep Research Agents:

Deep research agents are **advanced AI systems designed to handle complex, open-ended research by planning autonomously, using tools iteratively, and maintaining extended memory**. Unlike standard chatbots that provide instant answers, these agents take minutes or hours to methodically browse the web, verify sources, and synthesize their findings into cited reports.  Medium +1

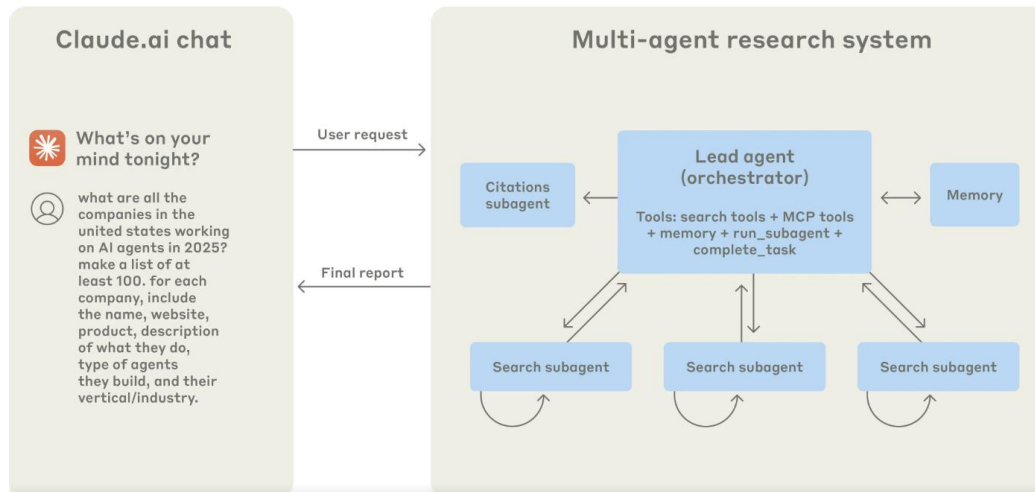
Why Are They Different?

While traditional "shallow" AI agents often struggle when tasked with long, multi-step problems, deep research agents break through these limitations by using four core components:  YouTube · LangChain +1

- **Autonomous Planning:** They generate dynamic to-do lists, break down broad questions into manageable sub-tasks, and adjust their strategy on the fly based on what they discover.  YouTube · LangChain +1
- **Sub-agent Delegation:** Instead of relying on a single "jack-of-all-trades" model, the main agent can spawn specialized sub-agents (e.g., a dedicated researcher or a critique agent) to explore specific angles simultaneously.  YouTube · LangChain +1
- **Persistent File Systems:** They use a virtual file space or long-term memory to collaborate, share context, and document partial findings without losing track over extended execution windows.  LangChain +1
- **Evaluator Feedback Loops:** They continuously self-correct by having one AI model generate an answer while a secondary AI or an evaluator checks it for accuracy, restarting the search if the information is incomplete or incorrect.  AnswerRocket

1. **Query Understanding and Planning:** Upon receiving a research query, the agent often starts by clarifying the scope and objectives with the user. It then formulates a plan, breaking down the research task into smaller, manageable steps.
2. **Information Gathering (Tool Use):** The agent autonomously uses tools, primarily web browsers, to search for relevant information online. It may access dozens or even hundreds of sources.
3. **Analysis and Synthesis:** The agent reads and analyzes the gathered information, extracting key points, identifying patterns, comparing data across sources, and potentially using other tools (like code interpreters for data analysis) to process the findings.
4. **Iterative Refinement:** The process is often iterative. Based on initial findings, the agent might refine its search queries, seek out additional sources, or revisit previous steps to deepen its understanding or resolve conflicting information.
5. **Report Generation:** Finally, the agent synthesizes its findings into a structured, often well-cited report, presenting the information in a coherent and organized manner.

High-level Architecture of Advanced Research



Example of a Deep Research Agent:

Prompt: How would I go about designing a Deep research Agent system that answers questions on diabetes. The system would consult reputed publications, especially the recent papers, published books, and other reputed resources on the subject on the Internet.

Here's how to design a **Deep Research Agent system for Diabetes** using CrewAI:

Recommended Architecture

Use a **Hierarchical Process** with a manager agent coordinating specialized sub-agents:

Agents

1. **Manager Agent**— orchestrates the crew, delegates tasks, validates outputs
2. **Literature Search Agent**— finds recent academic papers (arXiv, PubMed-style searches)
3. **Web Research Agent**— searches reputed web sources (journals, health organizations)
4. **Synthesis Agent**— compiles and summarizes findings with citations

Key Tools to Use

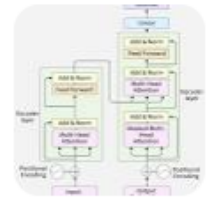
Tool	Use
ArxivPaperTool	Find recent academic papers without an API key.
TavilyResearchTool	Conduct multi-step web research and produce cited reports.
SerperDevTool	Search Google for reputable publications.
ScrapeWebsiteTool	Extract content from selected journal or book pages.
PDFSearchTool	Search downloaded PDFs using RAG.

7. What is a Transformer model:

AI Overview



A **transformer model** is a revolutionary deep-learning neural network introduced in 2017 that powers modern generative AI. It processes sequential data (like words in a sentence) all at once rather than one by one, using an "attention" mechanism to understand how distant words relate to each other. [NVIDIA Blog +1](#)



How it Works

Unlike older models (like RNNs) that analyzed text in strict order, transformers rely on core building blocks to grasp meaning: [🔗](#)

- **Self-Attention:** This is the magic of the transformer. The model calculates how much weight or "attention" it should assign to every word in a text sequence relative to every other word. This allows it to figure out context. For example, in the sentence "The bank of the river," self-attention helps the model understand that "bank" means the side of a river, not a financial institution, by linking the word to "river". [NVIDIA Blog +3](#)
- **Parallel Processing:** Because it doesn't need to read words sequentially, a transformer can process an entire book or dataset simultaneously, massively speeding up training and execution times. [Coursera +1](#)
- **Encoder-Decoder Architecture:** Traditional transformers are split into two parts. The **encoder** reads and compresses the input data, and the **decoder** translates it to generate the output. [Coursera +1](#)

Example:

The animal didn't cross the street because it was too tired.--> it is related to animal
The animal didn't cross the street because it was too wide.--> it is related to street

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